A BIRATIONAL PROPER MORPHISM ONTO A NOETHERIAN ALGEBRAIC SPACE IS AN ISOMORPHISM IN CODIMENSION ONE AT NORMAL POINTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-over-fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the dense open isomorphism for part (1), the reduction of the codimension one case to the finite case, the neighbourhood on which the morphism becomes finite, and the etale localization to affine schemes reducing to the case of schemes.

Editorial note. The label lemma-modification-normal-iso-over-codimension-1 names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source spaces-over-fields.tex –label lemma-modification-normal-iso-over-codimension-1 –start 373 –end 416 –local-id 28059 returned RESULT=OK (L28059 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-spaces-over-fields/candidates-halfA.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Isomorphism over a dense open for birational morphisms. Decent Spaces, the lemma that a birational morphism of Noetherian spaces is an isomorphism over a dense open, together with the fact that Noetherian algebraic spaces are decent. Supplies part (1) and the reduction to the codimension one statement.
- Finiteness of a proper morphism with finite fibre over a point. Cohomology of Spaces, the lemma making a proper morphism finite in a neighbourhood of a point with finite fibre, and Decent Spaces, the lemma verifying the conditions on the fibre. Supplies the reduction to the case where the morphism is finite.
- The scheme case of the statement. Varieties, the lemma treating a finite birational morphism of Noetherian affine schemes at a point whose local ring is a normal domain. Supplies the conclusion after reducing to affine schemes.
- Etale localization of the property of being an isomorphism over an open. Same chapter, the lemma that finiteness in codimension one reduces the codimension one case to the finite case, and Spaces, the lemma descending

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a property of representable morphisms from an etale cover. Supply the reduction of part (2) to part (3) and the passage from an open neighbourhood on an etale cover to an open neighbourhood in the space.

Source reference. Original lemma 0B4D: sources/stacks-chow/tags.txt, source lines 13996–13996. Prerequisite pointer invoked at proof line(s) 410; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0BBF: sources/stacks-chow/tags.txt, source lines 14243–14243. Prerequisite pointer invoked at proof line(s) 393; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0ACK: sources/stacks-chow/tags.txt, source lines 13058–13058. Prerequisite pointer invoked at proof line(s) 402; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02MA: sources/stacks-spaces-over-fields/pinned-original-source.tex, source lines 341–341. Prerequisite pointer invoked at proof line(s) 396; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0A4W: sources/stacks-chow/tags.txt, source lines 12789–12789. Prerequisite pointer invoked at proof line(s) 400; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03KD: sources/stacks-chow/tags.txt, source lines 4673–4673. Prerequisite pointer invoked at proof line(s) 408; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 0BBL: sources/stacks-chow/tags.txt, source lines 14249–14249. Prerequisite pointer invoked at proof line(s) 384; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0BFP: sources/stacks-chow/tags.txt, source lines 14392–14392. Prerequisite pointer invoked at proof line(s) 415; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let X be a Noetherian algebraic space over S. Let $f : Y \to X$ be a birational proper morphism of algebraic spaces with Y reduced. Let $U \subset X$ be the maximal open over which f is an isomorphism. Then U contains*

(1) *every point of codimension 0 in X,*
(2) *every $x \in |X|$ of codimension 1 on X such that the local ring of X at x is normal (Properties of Spaces, Remark 0BBL), and*
(3) *every $x \in |X|$ such that the fibre of $|Y| \to |X|$ over x is finite and such that the local ring of X at x is normal.*

Proof. Part (1) follows from Decent Spaces, Lemma 0BBF (and the fact that the Noetherian algebraic spaces X and Y are quasi-separated and hence decent). Part (2) follows from part (3) and Lemma 02MA (and the fact that finite morphisms have finite fibres). Let $x \in |X|$ be as in (3). By Cohomology of Spaces, Lemma

0A4W (which applies by Decent Spaces, Lemma 0ACK) we may assume f is finite. Choose an affine scheme X' and an étale morphism $X' \to X$ and a point $x' \in X$ mapping to x. It suffices to show there exists an open neighbourhood U' of $x' \in X'$ such that $Y \times_X X' \to X'$ is an isomorphism over U' (namely, then U contains the image of U' in X, see Spaces, Lemma 03KD). Then $Y \times_X X' \to X$ is a finite birational (Decent Spaces, Lemma 0B4D) morphism. Since a finite morphism is affine we reduce to the case of a finite birational morphism of Noetherian affine schemes $Y \to X$ and $x \in X$ such that $\mathcal{O}_{X,x}$ is a normal domain. This is treated in Varieties, Lemma 0BFP. □